Foundations of Computing Theory—Homework 7
Due: Wednesday, February 6, 2013, 2:00pm

- You only have to explain your answer if this is stated in the question.
- Before you start on the homework, please read the rules on collaboration and submission in the syllabus.
- Keep in mind that, as stated in the syllabus, I will stop answering homework questions at 5pm on Tuesday.

1. Consider the language $A = \{a^i b^i c^i \mid i \geq 0\}$.
   (a) Describe a TM T that decides A. Use the format and level of detail of the description of M_1 at the top of page 167.
   (b) Draw the state diagram of T.

2. A *queue automaton (QA)* is just like a PDA, but with a queue instead of a stack. The 6-tuple definition of a QA is exactly the same as the 6-tuple definition of a PDA (Definition 2.13). The transition $(q, c) \in \delta(p, a, b)$ (which in the state diagram shows as an arrow from p to q labeled $a, b \to c$) means that if the QA is in state p, reading input a, and the current symbol at the start of the queue is b, then the QA can change state to q, dequeue b (i.e., delete b from the front of the queue), and enqueue c (i.e., add c to the end of the queue). Recall that a, b, and c can be ϵ. Acceptance is defined in the same way as for a PDA.

   Draw the state diagram of a QA that accepts the language $\{a^i b^i c^i \mid i \geq 0\}$. You QA should not be overly complicated.

3. (a) Show that the collection of decidable languages is closed under concatenation. Use the format and level of detail of the answer to Problem 3.15(a).
   (b) Does your construction from part (a) also show that RE (the class of recursively enumerable (or Turing-recognizable) languages) is closed under concatenation? Explain your answer. (It may be helpful to look at the answers to Problems 3.15(a) and 3.16(a).)
   (c) Does your answer to part (b) imply that RE is not closed under concatenation? Briefly explain your answer.

4. Consider the problem of determining whether the language accepted by a DFA is non-trivial, i.e., given a DFA, accept if and only if the language accepted by the DFA is not the empty set and not Σ^*.
   (a) Formulate this problem as a language.
   (b) Show that your language from part (a) is decidable. For your answer, please give the high-level description of a Turing machine that decides your language using Sipser notation (see Section 4.1), and briefly explain why your TM decides the language.